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Compatibility of Spectral and Operator Reductions of a First-Principles Silicon Hamiltonian

A constrained multi-objective formulation with aligned-operator diagnostics

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Content-slide evidence: charcoal/blue. Red: prospective work or an unexecuted extension.

Long-term target: a substitutional-phosphorus operator

Host route

periodic DFT: Si $\longrightarrow$ selected Bloch subspace $\longrightarrow$ Wannier frame

$$\{H^{W,\mathrm{Si}}(\mathbf{R})\}.$$

Impurity route

periodic DFT: Si:P $\longrightarrow$ selected Bloch subspace $\longrightarrow$ Wannier frame

$$\{H^{W,\mathrm{Si:P}}(\mathbf{R})\}.$$

Reconcile supercell, rank, subspace, orbital ordering, geometry, energy zero, units, and gauge

$$\Delta H_P^W(\mathbf{R}) = \widetilde{H}^{W,\mathrm{Si:P}}(\mathbf{R}) - \widetilde{H}^{W,\mathrm{Si}}(\mathbf{R}).$$

Reduced-model target

$$H^W_{\mathrm{model}} = H^W_{\mathrm{host}} + \Delta H_P^W,$$

followed by withheld checks against the aligned Si:P calculation before any finite-range compression is accepted.

ICMSEP scope: establish a compatible bulk-Si reference. The production Si:P calculation and authenticated impurity perturbation are subsequent prospective work; the current silicon evidence is a bounded tutorial control.

Can the operator be extracted in any frame? Yes, covariantly

Fixed selected subspace and one common frame

For any periodic unitary $V(\mathbf{k})$ applied to both represented systems,

$$H_X^V(\mathbf{k}) = V^\dagger(\mathbf{k}) H_X(\mathbf{k}) V(\mathbf{k}),$$
$$\Delta H_P^V(\mathbf{k}) = V^\dagger(\mathbf{k}) \Delta H_P(\mathbf{k}) V(\mathbf{k}).$$

The spectra and abstract operator difference are unchanged. The matrix entries and their distribution among real-space blocks change covariantly.

one common frame $\Longrightarrow$ equivalent representations of one operator difference

Additional locality condition

A smooth periodic $V(\mathbf{k})$ is needed for a localized Wannier representation. A discontinuous or rapidly varying gauge is algebraically valid but can produce long-ranged blocks.

Can independently chosen frames be subtracted? No

Independent coordinate arrays cannot be subtracted directly

$$V_P^\dagger H_{\mathrm{Si:P}} V_P - V_0^\dagger H_{\mathrm{Si}} V_0$$

has no common-coordinate meaning unless an explicit, well-conditioned map identifies the two frames.

Gauge mismatch

Equal selected subspaces with a full-rank cross-overlap admit a unitary alignment. Phase, permutation, and internal unitary freedom can then be reconciled.

Subspace mismatch

Rank loss or unequal projectors is not removable by gauge alignment. The projection/disentanglement contract must be repaired or the residual reported.

For interpretable impurity blocks, prefer a shared projector-defined or host-anchored frame over two independently optimized MLWF gauges.

Is the Kohn–Sham solution unique? Specify the object

Map or object	Qualified uniqueness statement
$n_0 \mapsto v_{\mathrm{ext}}$	Hohenberg–Kohn: unique up to an additive constant under the standard nondegenerate ground-state assumptions [1].
$n_0 \mapsto v_s$	The local KS potential is unique up to a constant when the density is noninteracting v-representable and the relevant uniqueness assumptions hold.
$v_s \mapsto$ canonical KS states	Nondegenerate eigenstates are fixed only up to phase; degenerate eigenspaces admit unitary rotation.
occupied projector $P \mapsto$ frame F	Never unique: F and FU represent the same projector for every unitary U.

The apparent contradiction

A density may determine a qualified local potential without determining a unique orbital frame. “Invert the density to a potential” and “choose a basis inside an occupied or selected subspace” are different inverse problems.

Why orbital rotations can leave the density unchanged

Let $F(\mathbf{k})$ collect equally occupied Bloch orbitals. Then

$$P = FF^\dagger, \qquad F \mapsto FU, \qquad P \mapsto FUU^\dagger F^\dagger = P,$$

and consequently

$$n(\mathbf{r}) = \sum_{\mathbf{k}} \sum_{n \in \mathcal{S}} f_{\mathbf{k}} |\psi_{n\mathbf{k}}(\mathbf{r})|^2$$

is unchanged.

Wannier gauge freedom

The projector and equal-occupation density do not select an internal frame. A smooth periodic $U(\mathbf{k})$ selects localized Wannier representatives of the same subspace.

Restricted freedom

Mixing unequal occupations generally changes the density. Mixing nondegenerate eigenstates preserves the selected subspace, but the Hamiltonian becomes non-diagonal and the rotated orbitals are no longer individual KS eigenstates.

Where information is retained, quotiented out, or discarded

Map	What remains	What is no longer explicit
Many-body $\Psi \mapsto n$	Ground-state density	Correlations and phases; exact Hohenberg–Kohn theory gives conditional formal recovery, not an orbital gauge from the density alone [1]
KS frame $F \mapsto P = FF^\dagger$	Selected subspace and equal-occupation density	Internal unitary frame $U(\mathbf{k})$
Bloch matrix $H(\mathbf{k}) \mapsto \operatorname{spec} H(\mathbf{k})$	Eigenvalues	Eigenvectors, projectors, orbital matrix structure, and hopping distribution
$H(\mathbf{k}) \leftrightarrow \{T_{\mathbf{R}}\}$	Complete operator representation	Nothing in the complete transform; information is lost only by sampling, projection, or truncation

Practical consequence

A wavefunction file retains one gauge-dependent representative $F(\mathbf{k})$. Wannier construction uses those states and overlaps; the charge density alone is not sufficient to construct or align the desired tight-binding operator.

What physical change does the subtraction contain?

Let $\mathcal{R}_0$ be the pristine geometry, $\mathcal{R}_P$ the Si:P-relaxed positions, and $\widetilde{H}_X^W(\mathcal{R}) = \mathcal{A}_X[H_X^W(\mathcal{R})]$ the Hamiltonian after declared state-space, energy-reference, and frame alignment. Two endpoint calculations give

$$\Delta H^W_{P,\mathrm{total}} = \widetilde{H}^W_{\mathrm{Si:P}}(\mathcal{R}_P) - \widetilde{H}^W_{\mathrm{Si}}(\mathcal{R}_0),$$

which generally combines substitution chemistry, self-consistent electronic screening, and structural relaxation.

A matched-geometry Si calculation separates two channels

$$\begin{aligned}
\Delta H^W_{P,\mathrm{chem}} &= \widetilde{H}^W_{\mathrm{Si:P}}(\mathcal{R}_P) - \widetilde{H}^W_{\mathrm{Si}}(\mathcal{R}_P),\\
\Delta H^W_{P,\mathrm{relax}} &= \widetilde{H}^W_{\mathrm{Si}}(\mathcal{R}_P) - \widetilde{H}^W_{\mathrm{Si}}(\mathcal{R}_0),\\
\Delta H^W_{P,\mathrm{total}} &= \Delta H^W_{P,\mathrm{chem}} + \Delta H^W_{P,\mathrm{relax}}.
\end{aligned}$$

The production protocol must fix charge state, spin, supercell finite-size controls, structural branch, and whether the target is the screened Kohn–Sham perturbation or another effective operator.

Why four controls precede production subtraction

Production question

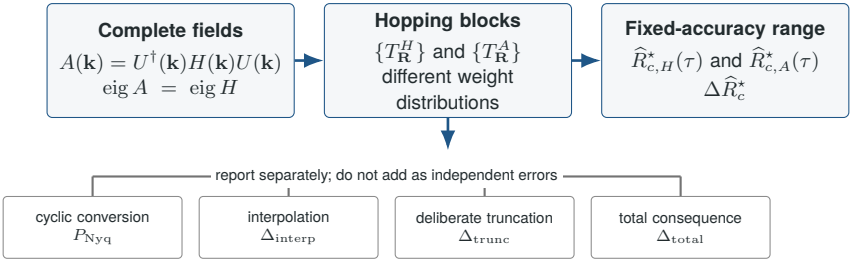
What must be controlled before independently generated Si and Si:P DFT/Wannier operators can be aligned, subtracted, and compressed without confusing gauge, discretization, or model-reduction effects with the impurity response?

1. **Exact 1-D rank-two benchmark:** isolates the gauge/locality mechanism and defines equal-accuracy range comparison.
2. **2-D spectral-to-TB control:** tests multidimensional state spaces, Fourier conversion, mixed hoppings, and interpolation.
3. **Rank-three multiband control:** tests pointwise versus restricted frame alignment and its locality consequences.
4. **Silicon tutorial bridge:** tests same-frame operator decomposition and component-sensitive truncation on wavefunction-derived data.

$$\{T_{\mathbf{R}}\} \longleftrightarrow H(\mathbf{k}) \longrightarrow \operatorname{spec} H(\mathbf{k})$$

Convention: $\mathbf{k}$ is the Bloch wave vector ($\hbar\mathbf{k}$ is crystal momentum); $\hat{\mathbf{p}} = -i\hbar\nabla$ is the canonical momentum operator.

Why spectral targets are necessary but not sufficient



G: state-space and transport gates **O**: representation/locality observables **C**: approximation consequences **V**: redundant verification of every arrow

The benchmark isolates representation-induced locality from projection, discretization, and material-model errors; it does not replace physical spectral targets with diagnostic quantities.

Bulk-Si reduction as a two-objective problem

For each declared tight-binding class $\mathcal{M}_j$, define two independent objectives

$$f_s(\theta) = d_s(\mathbf{s}_\theta, \mathbf{s}_{\mathrm{DFT}}),$$
$$f_o(\theta) = d_o\left(H^W_\theta, H^W_{\mathrm{DFT}}\right),$$

where $\mathbf{s}$ contains selected bands, gap, valley, and effective-mass observables, while f_o compares aligned localized operators.

MOO problem within model class $\mathcal{M}_j$

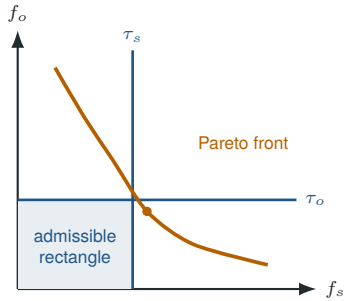
$$\underset{\theta\in\mathcal{M}_j}{\text{Pareto-minimize}} \quad \big(f_s(\theta), f_o(\theta)\big) \qquad \text{subject to qualification and model constraints.}$$

Do not hide the trade-off in one weighted score

A small weighted sum can compensate an unacceptable spectral error with a small operator error, or vice versa. The two objectives, their tolerances, and their Pareto trade-off remain visible.

Gap isolation, overlap rank, sewing, Parseval, and verification checks are constraints or evidence about the calculation—not replacement objectives.

Compatibility is a Pareto-feasibility question



$$\mathcal{A}_{s,j} = \{\theta : f_s(\theta) \le \tau_s\},$$
$$\mathcal{A}_{o,j} = \{\theta : f_o(\theta) \le \tau_o\}.$$

Compatible class

$$\mathcal{A}_{s,j} \cap \mathcal{A}_{o,j} \neq \varnothing.$$

A Pareto witness enters the admissible rectangle.

Model selection

Choose the smallest reviewed class j with a qualified common witness. Range or parameter count can be added as a declared third objective, but is not folded silently into f_s or f_o.

If both admissible sets are nonempty but disjoint, report their normalized operator-space separation rather than a weighted compromise: $\delta_j^* = \inf_{\theta_s \in \mathcal{A}_{s,j}, \theta_o \in \mathcal{A}_{o,j}} d_H(H_{\theta_s}, H_{\theta_o})$.

The production bulk-Si Pareto front and compatibility witness remain uncomputed.

Paired diagnostic ledger for the mechanism schematic

Code	Schematic attachment	Diagnostic families	Interpretation
G	complete fields and retained frame	external isolation gap; projector reconstruction; neighbor-overlap singular values; transport unitarity; reciprocal sewing and closure	must pass before observables are interpreted
O	restricted alignment and hopping blocks	$\mu_{\max}$; irreducible operator residual; block Hermiticity; nonsite and cumulative tail power	representation and locality observables
C	four lower channel boxes and fixed-accuracy range	Nyquist conversion; interpolation; deliberate truncation; total Hamiltonian and spectral consequences; first passing range	separately named, nonadditive consequences
V	every transformation arrow	Fourier nodal reconstruction; Parseval; Weyl bound; independent reconstruction; momentum/amplitude refinement	consistency evidence, not scientific acceptance

Traceability rule

The schematic supplies the object-and-transformation map; the ledger supplies the gate, observable, consequence, or check attached to each element. Neither is complete without the other.

Diagnostic roles: do not collapse them into one score

Qualification gates

- external isolation gap;
- overlap singular values;
- reciprocal sewing;
- Hermiticity.

Decide whether analysis may proceed.

Invariants

- complete bands;
- projectors;
- Wilson-phase multisets;
- covariant reconstruction.

Establish equivalence under the declared map.

Locality observables

- block powers;
- omitted tails;
- spread estimators;
- shell counts.

Measure representation structure.

Consequences

- interpolation;
- truncation;
- total errors;
- Weyl/Parseval;
- independent reconstruction.

Quantify and verify approximation effects.

A gate is not a convergence observable; omitted power is not a universal spectral bound; verification is not material validation.

Reciprocal sewing is a diagnostic ladder, not one number

Let $S_{\mathbf{G}}(\mathbf{k}) : \mathcal{H}_{\mathbf{k}} \to \mathcal{H}_{\mathbf{k}+\mathbf{G}}$ identify Bloch fibers across a reciprocal boundary.

$$
\begin{aligned}
H(\mathbf{k}+\mathbf{G})S_{\mathbf{G}} &= S_{\mathbf{G}}H(\mathbf{k}),\\
P(\mathbf{k}+\mathbf{G})S_{\mathbf{G}} &= S_{\mathbf{G}}P(\mathbf{k}),\\
F(\mathbf{k}+\mathbf{G}) &= S_{\mathbf{G}}F(\mathbf{k})W_{\mathbf{G}}(\mathbf{k}),\\
H_F(\mathbf{k}+\mathbf{G}) &= W_{\mathbf{G}}^{\dagger}H_F(\mathbf{k})W_{\mathbf{G}}.
\end{aligned}
$$

Level	Required diagnostics
Retained subspace	external isolation gap; constant rank; projector boundary covariance
Neighbor transport	overlap singular values; full-rank gate; Procrustes residual
Frame boundary	sewing-matrix unitarity; corrected endpoint discrepancy
Closed paths	closure/holonomy phases and branch-cut proximity
2-D/3-D extension	every primitive boundary; path consistency; plaquette closure; existence of one global periodic frame

Wilson-loop and plaquette phases are sewing diagnostics here, not topological claims.

1-D qualification and sewing: gates versus observations

E_{ref} is an arbitrary reference-energy normalization, not a band gap.

Diagnostic	Declared role or gate	Sampled observation
External isolation gap $\min_q(\epsilon_3 - \epsilon_2)/E_{\mathrm{ref}}$	must exceed 1.0	minimum 1.0566461
Neighbor/closure overlap $\sigma_{\min}$	must exceed 0.99; equality fails	minimum 0.9999989292
Neighbor Procrustes residual	conditioning record, not convergence observable	maximum 1.831×10^{-3}
Transport-unitarity discrepancy	frame tolerance 10^{-10}	maximum 1.89×10^{-15}
Frame-orthonormality discrepancy	frame tolerance 10^{-10}	maximum 3.34×10^{-15}
Corrected periodic-closure discrepancy	numerical sewing check	maximum 9.55×10^{-16}
Closure-holonomy phases	report ordered phase set; not a pass threshold	endpoint $(0.014181, 0.018807)$ rad

The gap and overlap thresholds are declared gates; 1.0566461 and 0.9999989292 are observed minima, not acceptance thresholds.

From sewn operator to finite-range model

Representation level	Diagnostic families	Scientific role
Retained Bloch operator	frame isometry; projector lift; parent–retained reconstruction; complete spectra	establish the operator being represented
Wannier/Fourier blocks	nodal round trip; Parseval; $T_{-\mathbf{R}} = T_{\mathbf{R}}^{\dagger}$; cyclic Nyquist power	verify conversion and ordinary-lattice structure
Cross-representation alignment	pointwise covariant residual; one-global-unitary residual; symmetry intertwiners; cross-boundary sewing compatibility	separate gauge recovery from restricted comparison
Locality	shell/block power; omitted power; spreads; tail norms; support-orbit closure	characterize representation, not spectral error
Approximation consequences	separate interpolation, truncation, and total Hamiltonian max/RMS errors; separate sorted-band errors; Weyl check	quantify what compression changes
Range decision	first passing sampled shell; predecessor failure; shifted evaluation and mesh refinement	equal-accuracy representation cost

1-D scientific observables: alignment and locality

$$\mu_{\max} = \frac{1}{\kappa} \max_{\phi} b_{\phi}, \qquad L^2_{\mathrm{op},*}(s,\lambda) = s^2 + \kappa(\lambda - \mu_{\max})^2 + \kappa(1 - \mu^2_{\max}),$$

$$\inf_{s\in\mathbb{R},\lambda\geq 0} L^2_{\mathrm{op},*} = \kappa\left[1 - \max(0, \mu_{\max})^2\right], \qquad P^X_{>R_c} = \frac{1}{2E^2_{\mathrm{ref}}} \sum_{|R|>R_c} \|T^X_R\|^2_F.$$

Quantity	Interpretation
$\mu_{\max}$	best continuously optimized global real-rotation correlation; an observable, not a transport gate
irreducible operator residual	matrix information inaccessible to the restricted global alignment although the complete fields remain pointwise unitarily equivalent
$P^X_{>0}$ and $P^X_{>R_c}$	non-onsite and cumulative omitted Frobenius power; locality diagnostics, not probabilities or universal spectral-error bounds
sampled range $\widehat{R}^{\star}_c(\tau)$	first integer cutoff passing a separately named Hamiltonian or band criterion; the immediately preceding cutoff must fail

1-D consequence channels and redundant checks

Channel/check	Compared objects	Sampled evidence
Cyclic Nyquist conversion	normalized power in the removed $-N/2$ representative	8.87×10^{-32} to 1.14×10^{-31}
Interpolation	finite interpolant versus direct field on disjoint mesh	max Hamiltonian 2.46×10^{-14}; band 2.35×10^{-14}
Deliberate truncation	truncated ordinary-lattice field versus full interpolant	matrix and band max/RMS retained separately
Total consequence	truncated field versus direct field	separate Hamiltonian and band cutoff decisions
Truncation Parseval	squared RMS Hamiltonian error versus normalized omitted power	discrepancies 8.33×10^{-17} and 1.11×10^{-16}
Weyl check	max band error divided by its Frobenius bound	ratio 0.0087 to $0.9984 \leq 1$
Momentum refinement	nested construction meshes plus shifted evaluation meshes	zero sampled cutoff-decision changes

Conversion, interpolation, truncation, and total consequences are nonadditive channels.

Material subtraction requires another diagnostic ladder

Level	Diagnostics and gates	Present status
DFT parents	structure and pseudopotential identity; SCF/cutoff/$\mathbf{k}$ convergence; complete target bands; symmetry and energy-reference conventions	production pending
Wannier subspaces	projectability; windows; disentanglement; rank; centers/spreads; interpolation; reciprocal sewing and symmetry covariance	tutorial only
Host–impurity identification	same supercell and orbital inventory; host lift; cross-overlap singular values; site/orbital matching; cross-system sewing and symmetry intertwiners	unexecuted
Aligned subtraction	common energy zero and units; pointwise/common-frame residuals; Hermiticity; exact readdition $H^W_{\mathrm{Si}} + \Delta H^W_P = H^W_{\mathrm{Si:P}}$	unexecuted
Physical decomposition	matched-geometry chemical and relaxation channels; charge/spin branch; supercell finite-size and far-field recovery	open controls
Reduced impurity model	locality tails; symmetry-resolved blocks; truncation channels; withheld bands, valley, masses, and operator residuals	unexecuted
Evidence identity	inputs, executable, pseudopotentials, source revision, checksums, verifier, and immutable evidence boundary	production pending

A later diagnostic cannot repair a failed state-space, sewing, or provenance gate.

Comparison contract: align spaces before subtraction

Parent and retained spaces

$$F(\mathbf{k}) : \mathbb{C}^r \to \mathcal{H}_p, \qquad F^\dagger F = I_r,$$

$$H_{\mathrm{ret}}(\mathbf{k}) = F^\dagger(\mathbf{k}) H_p(\mathbf{k}) F(\mathbf{k}).$$

A parent-space comparison requires the projector $P = FF^\dagger$ and an explicit lift $F H_{\mathrm{ret}} F^\dagger$.

Subtraction is undefined unless we fix

- rank and state-space map;
- basis and orbital ordering;
- gauge or admissible alignment;
- energy zero and units;
- lattice geometry and Fourier convention.

An eigenvalue array is spectral data; it is not a Hamiltonian in orbital coordinates.

Result 1 theory: exact rank-two geometry

Write the retained field as $H = \bar{E} I_2 + D$ with $\operatorname{tr} D = 0$, and consider the positive trace–traceless candidate

$$C_{s,\lambda} = \bar{E} I_2 + s E_{\mathrm{ref}} I_2 + \lambda D, \qquad \lambda > 0.$$

Under the declared weighting and one global real rotation,

$$L^2_{\mathrm{spec}}(s, \lambda) = s^2 + \kappa(\lambda - 1)^2,$$
$$L^2_{\mathrm{op},*}(s, \lambda) = s^2 + \kappa(\lambda - \mu_{\max})^2 + \kappa(1 - \mu^2_{\max}).$$

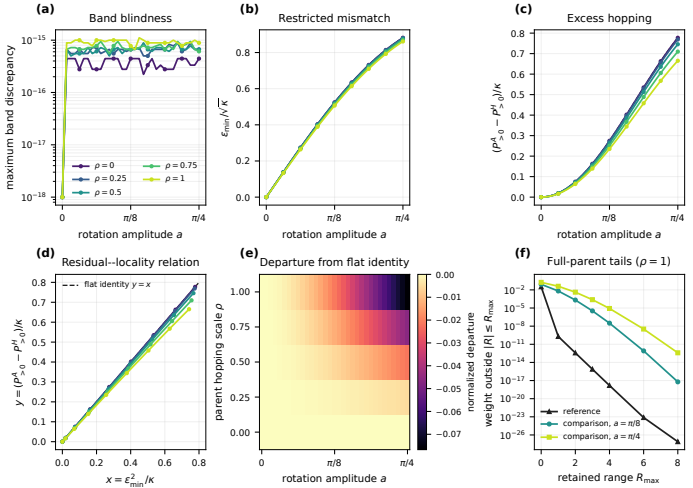
Smooth flat parent

$\mu_{\max}(a) = J_0(2a)$ gives exact alignment-threshold laws.

Exact locality identity

The irreducible squared matrix residual equals normalized Fourier power outside the onsite block.

Result 1 numerical outcome: equal-accuracy range inflation



- Complete rank-two fields are related by a smooth $\mathbf{k}$-dependent $SO(2)$ rotation.
- Their bands agree pointwise.
- One global real rotation cannot recover the full gauge.
- At $\tau = 10^{-8} E_{\mathrm{ref}}$:

$$\widehat{R}^{\star}_{c,\mathrm{ref}} = 4, \qquad \widehat{R}^{\star}_{c,\mathrm{comp}} = 10.$$

Interpretation

$\Delta \widehat{R}^{\star}_{c} = 6$ measures sampled representation cost, not physical inequivalence.

Result 2 construction: spectral data to tight binding

For $d \in \{2, 3\}$, keep four orbitals per cell and place the tensor structure in the lattice translations:

$$\widehat{K}^{(d)}_{\rho,N} = I_{N^d} \otimes T_0 + \frac{\rho}{d} \sum_{\alpha=1}^{d} \left(\mathsf{S}_{\alpha,N} \otimes T_+ + \mathsf{S}^{\dagger}_{\alpha,N} \otimes T_- \right).$$

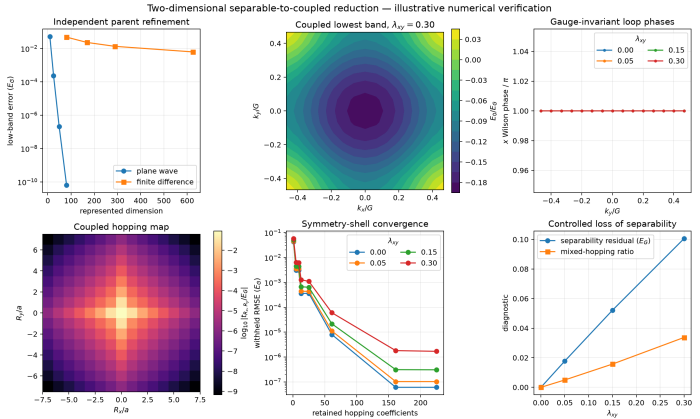
cell–orbital operator $\xleftrightarrow{\mathcal{F}_{d,N}}$ Bloch direct sum $\xleftrightarrow{Q(\mathbf{k})}$ complete eigensystem

$\xrightarrow{F(\mathbf{k})}$ retained Bloch field $\xleftrightarrow{\mathcal{F}^{-1}_{d,N}}$ Wannier/TB blocks

Projection is not structural compression

Even a gapped finite-range parent can acquire infinite Fourier support after spectral projection because its projector need not be a finite Fourier polynomial.

Result 2 evidence: reconstruction versus interpolation



- At zero mixed coupling, the represented Kronecker-sum discrepancy is zero.
- Product-spectrum discrepancy: $3.55 \times 10^{-14} E_{\mathrm{ref}}$.
- At $\lambda_{xy} = 0.30$, mixed-direction hopping norm: $1.94 \times 10^{-3} E_{\mathrm{ref}}$.
- Complete 15×15 transform reconstructs its nodes within $1.11 \times 10^{-15} E_{\mathrm{ref}}$.
- Independent 31×31 evaluation retains nonzero interpolation error.

Exact sampled reconstruction $\neq$ off-node convergence.

Result 3: pointwise and global unitary alignment

For equal-rank frames, form the cross-overlap

$$M(\mathbf{k}) = F^{\dagger}_{\mathrm{cand}}(\mathbf{k})F_{\mathrm{ref}}(\mathbf{k}) = L(\mathbf{k})\Sigma(\mathbf{k})V^{\dagger}(\mathbf{k}).$$

Pointwise family

$$U(\mathbf{k}) = L(\mathbf{k})V^{\dagger}(\mathbf{k}).$$

Align each $\mathbf{k}$ point before Fourier transformation. This can mix hopping ranges.

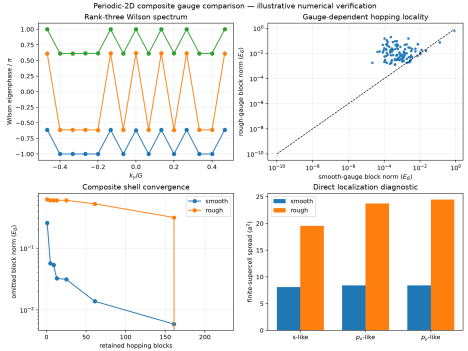
One-global-unitary family

$$B = \sum_{\mathbf{k}} M(\mathbf{k}) = L_B\Sigma_B V_B^{\dagger}, \qquad U_{\mathrm{glob}} = L_B V_B^{\dagger}.$$

One coordinate identification is used for the entire Brillouin zone.

These are different scientific restrictions, not two approximations to the same fit [2, 3]. Rank deficiency—including equality at the cutoff— fails closed.

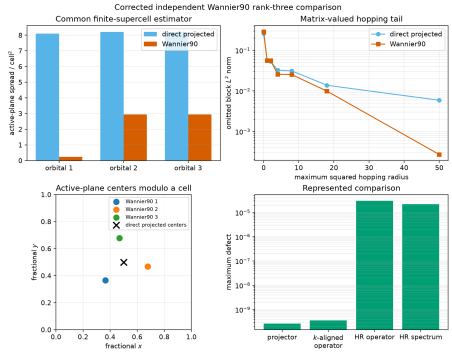
Result 3a: multiband gauge consistency does not fix locality



- Lowest-three-band smooth and rough frames.
- Spectral agreement: $1.22 \times 10^{-15} E_{\mathrm{ref}}$.
- Wilson-phase multiset agreement: 2.22×10^{-15}.
- Common-grid spreads: $24.90a^2$ versus $67.70a^2$.
- Tail beyond squared-radius 18: $1.38 \times 10^{-2} E_{\mathrm{ref}}$ versus $5.25 \times 10^{-1} E_{\mathrm{ref}}$.

Wilson phases are sewing diagnostics here, not a continuum topological claim.

Result 3b: one algorithm-selected Wannier90 frame



- Synthetic 2-D, rank-three subspace; no disentanglement.
- Projector discrepancy: 2.65×10^{-10}.
- Pointwise aligned operator discrepancy: $3.57 \times 10^{-10} E_{\mathrm{ref}}$.
- Restricted translation/global-unitary family leaves $1.490 E_{\mathrm{ref}}$.
- Radius-50 tails: $2.72 \times 10^{-4} E_{\mathrm{ref}}$ and $5.90 \times 10^{-3} E_{\mathrm{ref}}$.

One finite algorithm-selected comparison—not global MLWF optimality [4, 5].

Periodic DFT still produces Bloch states

After self-consistency, a periodic Kohn–Sham operator commutes with lattice translations, so its eigenstates may be written [6]

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}), \qquad u_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{G}} C_{n\mathbf{k}}(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}}.$$

What $\mathbf{k}$ means

$\mathbf{k}$ labels a translation character; $\hbar\mathbf{k}$ is crystal momentum modulo a reciprocal vector. It is not generally an eigenvalue of $\hat{\mathbf{p}}$.

What $\hat{\mathbf{p}}$ means

On a plane-wave component,

$$\hat{\mathbf{p}}\, e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} = \hbar(\mathbf{k}+\mathbf{G}) e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}.$$

A Bloch state generally contains many such components.

The theorem also accommodates lattice-periodic nonlocal pseudopotential operators; it is not restricted to a local scalar potential.

Wavefunction files serialize familiar Bloch states

Typically represented

- finite-cutoff coefficients $C_{n\mathbf{k}}(\mathbf{G})$;
- a finite $\mathbf{k}$ mesh, bands, spins or spinors;
- eigenvalues, occupations, and implementation metadata.

VASP calls one such file `WAVECAR`; Quantum ESPRESSO uses implementation- specific wavefunction files such as `.wfc*`.

Not fixed by that file alone

- a canonical band phase or multiband gauge;
- a Wannier subspace or localization choice;
- the full self-consistent density and potential;
- pseudopotential/PAW augmentation data;
- energy-reference and provenance contracts.

wavefunction files + companion data $\rightarrow \{\psi_{n\mathbf{k}}\} \rightarrow$ subspace/projectors $\rightarrow U(\mathbf{k}) \rightarrow$ Wannier/TB blocks.

The gauge step is essential: within an isolated or selected subspace, $\psi_{n\mathbf{k}} \mapsto \sum_m \psi_{m\mathbf{k}} U_{mn}(\mathbf{k})$ preserves the represented subspace but can change locality [4, 5].

Wannier transformation: a localized operator representation

For a selected rank-r Bloch subspace and gauge $U(\mathbf{k})$,

$$|\mathbf{R}n\rangle = \frac{1}{\sqrt{N_k}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \sum_{m=1}^{r} |\psi_{m\mathbf{k}}\rangle U_{mn}(\mathbf{k}).$$

A translation-covariant operator is represented by real-space blocks

$$O^W_{mn}(\mathbf{R}) = \langle \mathbf{0}m|\hat{O}|\mathbf{R}n\rangle = \frac{1}{N_k} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \left[U^\dagger(\mathbf{k})O(\mathbf{k})U(\mathbf{k})\right]_{mn}.$$

What localization can provide

A nonlocal operator kernel can have rapidly decaying matrix elements in a localized Wannier basis. This exposes a physically useful local representation and possible finite-range compression.

What it does not guarantee

The operator has not become local as an abstract operator, and its Wannier blocks need not have finite support. Decay depends on the subspace, gauge, regularity, gaps, and possible topological obstruction.

Eigenvalues plus $U(\mathbf{k})$ determine the projected Hamiltonian; kinetic, pseudopotential, and other components require their own same-frame matrix elements.

Result 4: silicon tutorial operators in one common frame

$$
\begin{aligned}
W(\mathbf{k}) &= U_{\mathrm{dis}}(\mathbf{k})U(\mathbf{k}),\\
H^W(\mathbf{k}) &= W^\dagger \,\mathrm{diag}(\epsilon) W,\\
T^W(\mathbf{k}) &= W^\dagger C_{\mathrm{PW}}^\dagger \,\mathrm{diag}\left(\frac{|\mathbf{k}+\mathbf{G}|^2}{2}\right) C_{\mathrm{PW}} W,\\
R^W(\mathbf{k}) &= H^W(\mathbf{k}) - T^W(\mathbf{k}).
\end{aligned}
$$

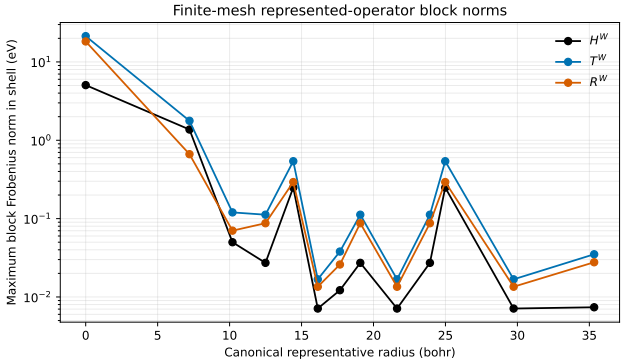
Finite checks

Same-frame subtraction discrepancy: 1.52×10^{-15} eV.
Native `_hr.dat` route difference: 1.28×10^{-5} eV.

Boundary

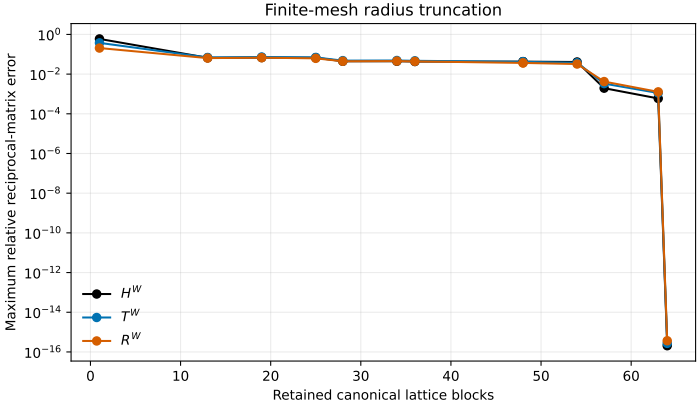
QE 7.5/Wannier90 3.1.0 tutorial, 25 Ry, 12 bands, four Wannier functions, 4^3 mesh. Not a converged production silicon model.

Result 4a: component block distributions differ



- At every nonzero represented radius, the maximum kinetic block exceeds the maximum remainder block by approximately 1.24–2.67.
- Cancellation can make the total block smaller than either component.
- Sixty-four nonmonotone blocks do not determine an infinite-lattice decay law.

Result 4b: 57-block truncation is not component-neutral



Maximum band error

Component	57/64 blocks
H^W	20.7 meV
T^W	84.3 meV
R^W	63.6 meV

All 64 blocks recover the sampled matrices to roundoff.

Failure of one finite-mesh truncation $\neq$ failure of tight binding generally.

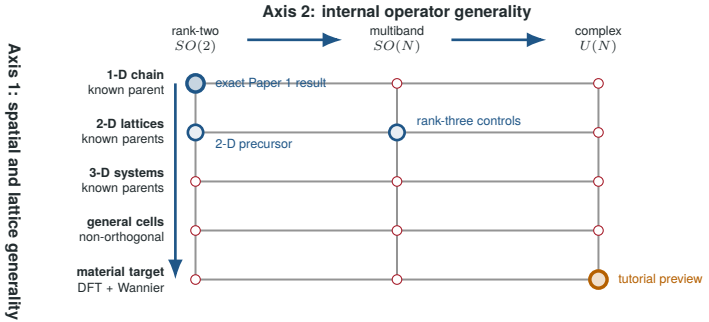
What the four results establish together

Stage	Established	Not established
Rank-two 1-D	Exact band equivalence with sampled range inflation	Universal range law
Scalar 2-D	Separation of tensor reconstruction, mixed coupling, and interpolation	3-D or non-orthogonal outcome
Rank-three 2-D	Gauge-consistent spectra with different finite locality diagnostics	Global localization optimum or topology
Silicon tutorial	Same-frame decomposition and failure of one component truncation	Production silicon adequacy

Common methodological conclusion

Complete fields, eigenvalues, projector/frame choices, Fourier conversion, and finite-range truncation must be kept as distinct information channels.

Two-axis validation ladder



Filled blue: established or bounded controlled evidence. Orange: bounded tutorial preview.

Red outlines: prospective experiments, not results.

This map separates accumulated bounded evidence from the unexecuted validation roadmap.

Why bulk compatibility matters for the Si:P objective

1. **Band equivalence does not determine hopping locality.**
2. **The desired Si:P perturbation is a difference of aligned operators,** not a difference of band energies or independently gauged Wannier files.
3. **Projection, gauge selection, Fourier conversion, interpolation, and deliberate truncation preserve different information.**
4. **Equal sampled accuracy—not equal cutoff—is the meaningful representation-cost comparison.**
5. **The silicon tutorial establishes failure of one component-wise finite-mesh truncation, not failure of tight binding generally.**

Next ICMSEP production step: a qualified bulk-Si parent, validated Wannier operator, and frozen compatibility search. Si:P follows only after that reference is accepted.

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Thank you

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